Polynomial Arithmetic

Adding, subtracting, and multiplying higher-degree polynomials

The model to work from

A product of two polynomials is the total area of a rectangle whose sides are split into the terms. Every cell of the box is one partial product, and adding the cells — collecting like terms as you go — gives the answer. Here is $(x+2)(x^2+3x)$ laid out that way:

$\times$	x^2	$3x$
x	x^3	$3x^2$
2	$2x^2$	$6x$

$\longrightarrow \quad x^3 + 5x^2 + 6x$

- Write every answer in standard form: highest power first, no like terms left.
- A subtraction sign in front of a bracket changes the sign of *every* term inside it.
- Where a box is drawn for you, fill it in and then collect the cells.

1. Simplify: $(3x^3 - 5x^2 + 2) + (x^3 + 4x^2 - 7x)$

Answer: ____________________

2. Simplify: $(6x^4 + 2x^2 - 9) - (2x^4 - 5x^2 + 3)$

Answer: ____________________

3. Multiply $(x+4)(x^2-3x+5)$. Fill in the box, then collect the cells.

$\times$	x^2	$-3x$	5
x			
4			

Answer: ___________________

4. Simplify: $2(x^3-4x)-3(2x^3+x-6)$

Answer: ___________________

5. Multiply $(2x^2-x+3)(x^2+5)$. Fill in the box, then collect the cells.

$\times$	$2x^2$	$-x$	3
x^2			
5			

Answer: ___________________

6. Consider the product $(x-2)(x^2+2x+4)$.

(a) Expand it and write the result in standard form.

Answer: ________________

(b) Substitute $x = 3$ into your expanded answer. What value do you get?

Answer: ________________

7. Simplify: $(x^4 - 3x^3 + x) - (4x^3 - 2x^2 + x - 6)$

Answer: ________________

8. Multiply $(x^2 + 3x - 1)(x^2 - 3x + 1)$. Fill in the box, then collect the cells. Watch the signs in the second row.

$\times$	x^2	$3x$	-1
x^2			
$-3x$			
1			

Answer: ________________

9. A rectangular garden bed measures $(2x + 5)$ metres along one side and $(x^2 - x + 3)$ metres along the other. Write its area as a polynomial in standard form.

Answer: ________________ m^2

10. Consider $(x^2 - 2x + 1)(x + 3) - (x^3 + x^2 - 5)$.

(a) Expand and simplify.

Answer: ____________________

(b) The product in front has degree 3, yet your answer does not. Explain how the degree of a product relates to the degrees of its factors, and why a subtraction can leave a result of lower degree.

Answer: ____________________